DATE: __/__/20__

Assignment 1.

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Reg# F24BSCS039
Course: Object Oriented Programming
Instructer: Dr. Huma Israr
Department & Program: IET, BSCS 2nd semester Morning
Date: 03/12/2025

Problem: Write A program in Java to multiply 2x2 matrixes using for, while, do-while & for each loop. For loop program should dry run as well.

Solution:

Using
While Loop

```
/*
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*/

public class While
{
public static void main (String [] args)
{

int matrix_a [][] = {{1,2},{3,4}};
```

DATE: __/__/20__

```
int matrix_b[][] = {{2,3},{4,4}};
// to store result
int result[][] = {{0,0},{0,0}};


// for rows
int i=0;
while (i<2)
{
    int j=0
    // for columns
    while (j<2)
    {
        int k=0;
        // extra loop, we need as per formula
        while (k<2)
        {
            result[i][j]= result[i][j] + (matrix_a[
                [i][k] * matrix_b[k][j]); // this
                will calculate the multiplication
        k++;} // end of loop k
        System.out.print(result[i][j] +" "); // this
            shows the result on screen
        j++;
    } // end of each columns
    System.out.print("\n"); moves cursor to
        next line after each row
    i++;
} // end of rows
}
} // end of program
```

DATE: __/__/20__

Using

Do-while Loop

```
/*
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*/
public Class Dowhile loop
{
public static void main(String[] args)
{

    int[][] matrix_a = {{1,2},{3,4}};
    int[][] matrix_b = {{2,3},{4,4}};
    int[][] result = {{0,0},{0,0}};

    int i=0;
    do // rows
    {
        ~~int~~ int j=0;
        do // columns
        {
            int k=0;
            do
            {
                result[i][j] = result[i][j] + (matrix_a[i][k]
                    * matrix_b[k][j]); // stores result
                k++;
            } while (k<2); // end of k loop
            System.out.print(result[i][j] + " ");
            j++;
        } while (j<2); // end of j columns
        System.out.print("\n"); // move to next line each row
        i++;
```

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DATE: __/__/20__

```
        i++;
    } while (i<2); // end of rows

}
} // end of program
```

Using for-Each/Enhenced for loop

Using for each loop is not suitable for matrix multiplication directly (I tried multiple times)

```
/*
    Name: sumama
    Reg# F24BSCS039
*/
public class Foreach
{
static public void main (String[] arguments)
{

    int[][] matrix_a = {{1,2},{3,4}}; // Matrix A
    int[][] matrix_b = {{2,3},{4,4}}; // Matrix B
    int[][] result = new int[2][2];
    // for multiplying we still need loop
      other than for each
    for (int i=0; i<2; i++)
    { // rows
        for (int j=0; j<2; j++)
        { // columns
            for (int k=0; k<2; k++) // k loop
            {
            result int result[][] = {{0,0},{0,0}};
```

DATE: __/__/20__

```
            result[i][j] += matrix_a[i][k] * matrix_b
                                [k][j]; // result calculat...
        } // end of k loop
    } // end of columns
} // end of rows

// Now showing result using for-each
   OR enhenced for loop
// for 2-D array, we need nested for each
    for (int row : result)
    {
        for (int res : row)
        {
        System.out.print(res + " ");
        }
    }

}
} // end of program
```

Using For Loop

```
/*
    Name: Sumama Rahim
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*/
public class Forloop
{
static public void main(String args[])
{
```

DATE: __/__/20__

```
int[][] matrix_a = {{1,2}, {3,4}};
int[][] matrix_b = {{2,3}, {4,4}};
int[][] result = new int[2][2];

for (int i=0 ; i<2 ; i++)
 {
    for (int j=0 ; j<2 ; j++)
    {
        for (int k=0; k<2 ; k++)
        {
            // result calculation & stores it here
            result[i][j] += matrix_a[i][k] * matrix
                _b[k][j];
        }// end of k loop
        // displaying the matrix (resultant) on
            screen
        System.out.print(result[i][j] +" ");
    }// end of columns
    // moving to next line after each row
    System.out.print("\n");
 }// end of rows

}
}// end of program
```

DATE: ___/___/20___

DRY RUN OF FOR LOOP PROGRAM

Result[i][j] += matrix_a[i][k] * matrix_b[k][j]	matrix_a[i][k]	matrix_b[k][j]	result[i][j]	O/P
= 0 + 2	0 0	00	00	2
= 2 + 2x4	0 1	10	00	$\boxed{10}$
= 0 + 1x3	0 0	01	01	3
= 3 + 2x4	0 1	11	01	$\boxed{11}$
= 0 + 3x2	1 0	00	10	6
= 6 + 4x4	1 1	10	10	$\boxed{22}$
= 0 + ~~9x9~~ 3x3	1 0	01	11	9
= 9 + 4x4	1 1	11	11	$\boxed{25}$

matrix_a[i][j] x matrix_b[k][j] = result[i][j]

$$\begin{array}{c} \begin{array}{cc} 00 & 01 \end{array} \\ \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \\ \begin{array}{cc} 10 & 11 \end{array} \end{array} \times \begin{array}{c} \begin{array}{cc} 00 & 01 \end{array} \\ \begin{bmatrix} 2 & 3 \\ 4 & 4 \end{bmatrix} \\ \begin{array}{cc} 10 & 11 \end{array} \end{array} = \begin{array}{c} \begin{array}{cc} 00 & 01 \end{array} \\ \begin{bmatrix} 10 & 11 \\ 22 & 25 \end{bmatrix} \\ \begin{array}{cc} 10 & 11 \end{array} \end{array}$$

outputs on each effective iteration